

Real campus sensor state
(local env., freshness, network,
temporal & neighbour context)



Semantic expert demonstrations



BC initialization $\rightarrow \pi_{BC}$



KL-CMAPPO (proposed)
• PPO policy improvement
• KL anchor to π_{BC}
• event / MAE / AoI constraints



Final shared policy
(shield-free)



TX / SKIP



Server monitoring
(causal reconstruction / LOCF)